

Technical Document #62: Natural Language Processing - Comprehensive Overview

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Coreference resolution identifies all expressions that refer to the same entity in a text. It is essential for understanding document-level coherence.

Machine translation converts text from one language to another. Neural machine translation using encoder-decoder architectures has achieved human-level performance.

Question answering systems extract answers from a given context. They typically consist of a retriever component and a reader component.

Named entity recognition (NER) identifies and classifies named entities in text into predefined categories such as person names, organizations, and locations.

Sentiment analysis determines the emotional tone behind a body of text. It is widely used in social media monitoring and customer feedback analysis.

Tokenization is the process of breaking text into individual units called tokens. Subword tokenization methods like BPE handle out-of-vocabulary words gracefully.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Key Takeaways:

- Machine translation converts text from one language to another.
- Word embeddings represent words as dense vectors in continuous space.
- Tokenization is the process of breaking text into individual units called tokens.